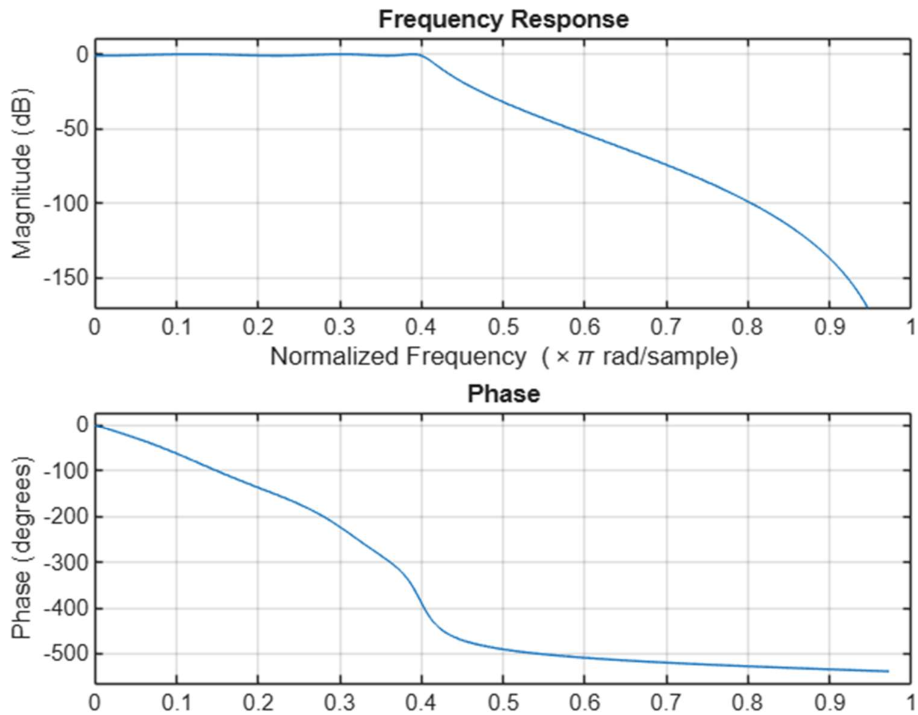
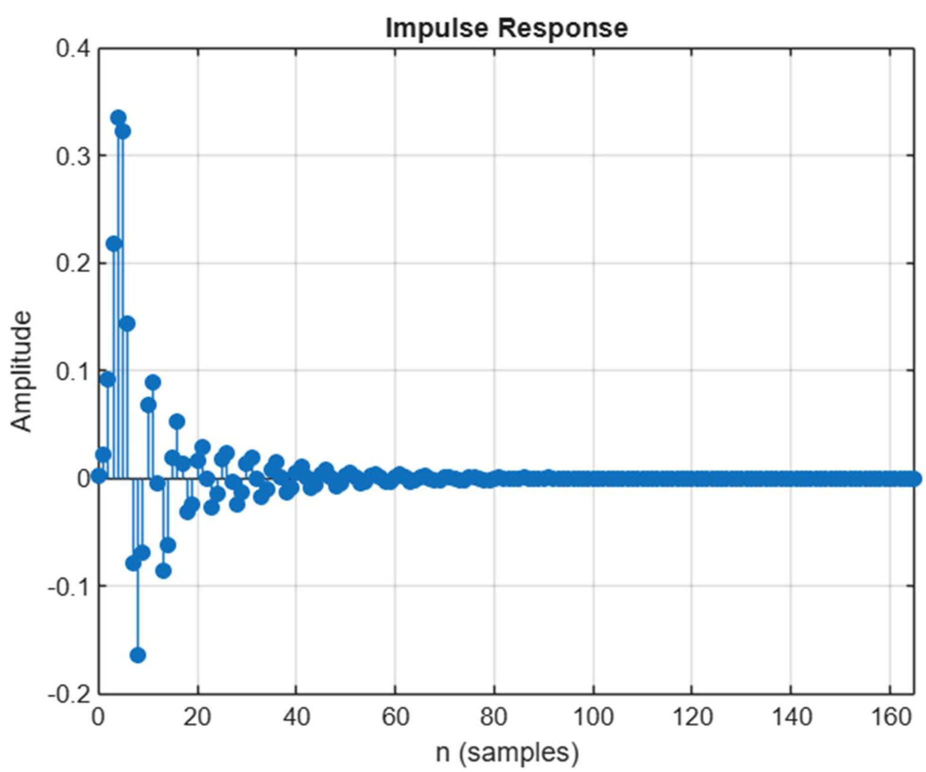
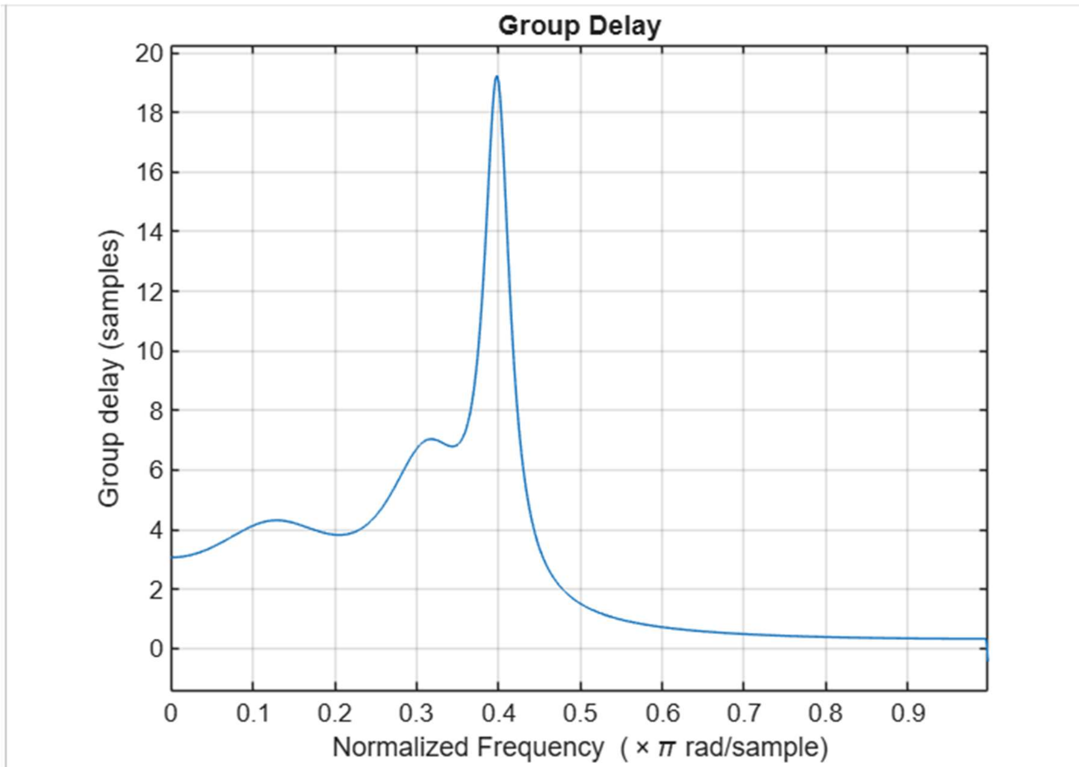
```
clc;
clear;
close all;

% Filter Specifications
Wp = 0.4;      % Passband Edge Frequency (normalized)
Ws = 0.55;     % Stopband Edge Frequency (normalized)
Rp = 1;        % Passband Ripple (dB)
As = 40;       % Stopband Attenuation (dB)

% Calculate Filter Order and Cutoff Frequency
[N, Wn] = cheb1ord(Wp, Ws, Rp, As);

% Design Chebyshev Type-I Low-Pass Filter
[b, a] = cheby1(N, Rp, Wn);

% Display Results
fprintf('Filter Order = %d\n', N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n', Wn);

% Stability Check
if all(abs(roots(a)) < 1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end

% Frequency Response
figure;
freqz(b, a);
title('Frequency Response');
```

```
% Impulse Response
figure;
impz(b, a);
title('Impulse Response');
grid on;

% Group Delay
figure;
grpdelay(b, a);
title('Group Delay');
grid on;

% Pole-Zero Plot
figure;
zplane(b, a);
title('Pole-Zero Plot');
```

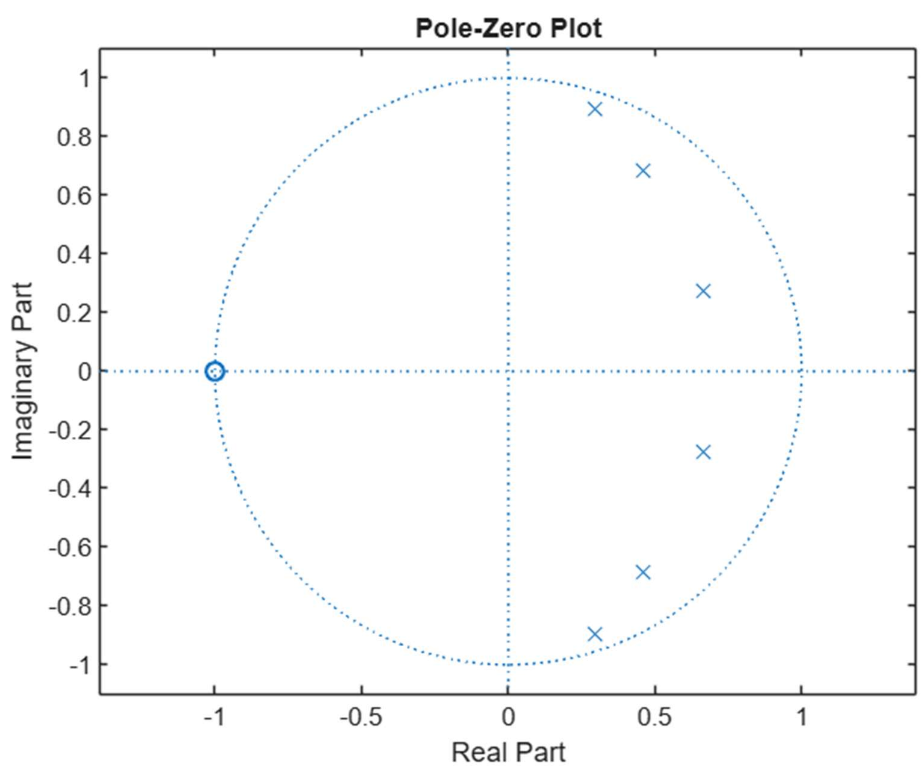
```
Filter Order = 6
Cutoff Frequency = 0.40 * pi rad/sample
System is STABLE
```







Group Delay
Group delay (samples)
Normalized Frequency (× π rad/sample)



Pole-Zero Plot
Imaginary Part
Real Part